Q.2. **[2.5+2.5=5 marks]**

Recall, Laplace distribution is given by

$$f_X(x|\mu,\sigma) = \frac{1}{\sqrt{2}\sigma} e^{-\frac{\sqrt{2}|x-\mu|}{\sigma}}, x \in \mathbb{R}$$

If data instances {-2, -1, 0, 2, 3} are obtained via sampling the above distribution, find the maximum likelihood estimate of (μ, σ).

Q. $$f_X(x/\mu,\sigma) = \frac{1}{\sqrt{2}\sigma} e^{-\frac{\sqrt{2}\,|x-\mu|}{\sigma}}$$

Sol → write the Likelihood function

Given n independent points $\{x_1, x_2, \dots x_n\}$, likelihood function is product of probability densities

$$\boxed{L(\mu,\sigma) = \prod_{i=1}^{n} f_X(x_i/\mu,\sigma)}$$

Substitute $f_X(x_i/\mu,\sigma)$

$$L(\mu,\sigma) = \prod_{i=1}^{n} \frac{1}{\sqrt{2}\sigma} e^{-\frac{\sqrt{2}\,|x-\mu|}{\sigma}}$$

$$L(\mu,\sigma) = \left(\frac{1}{\sqrt{2}\sigma}\right)^n \left(e^{-\frac{\sqrt{2}\,|x-\mu|}{\sigma}}\right)^n$$

Reference. { $\Pi \to$ Multiply. Hence, Power }

$$\boxed{L(\mu,\sigma) = \left(\frac{1}{\sqrt{2}\sigma}\right)^n \left(e^{-\frac{\sqrt{2}}{\sigma}\sum_{i=1}^{n}|x-\mu|}\right)}$$

Reference { with same base multiply, it will add up. $e^a \times e^b = e^{a+b}$ }

Sol → Log-likelihood function.

Reference { $\ln(A\times B) = \ln A + \ln B$ }

$$* \; l(\mu,\sigma) = \ln L(\mu,\sigma) = \ln\left[\underbrace{\left(\frac{1}{\sqrt{2}\sigma}\right)^n}_{A} \times \underbrace{\left(e^{-\frac{\sqrt{2}}{\sigma}\sum_{i=1}^{n}|x-\mu|}\right)}_{B}\right]$$

Reference { $\ln A^B = B\ln A$ }

$$* \; l(\mu,\sigma) = \ln\left(\frac{1}{\sqrt{2}\sigma}\right)^n + \ln\left(e^{-\frac{\sqrt{2}}{\sigma}\sum_{i=1}^{n}|x_i-\mu|}\right)$$

Reference { $\ln e^A = A$ }

$$* \; l(\mu,\sigma) = n\ln\left(\frac{1}{\sqrt{2}\sigma}\right) + \left(-\frac{\sqrt{2}}{\sigma}\sum_{i=1}^{n}|x_i-\mu|\right)$$

$$* \; \boxed{l(\mu,\sigma) = -n\ln(\sqrt{2}\sigma) - \frac{\sqrt{2}}{\sigma}\sum_{i=1}^{n}|x_i-\mu|}$$

Reference { $\ln\left(\frac{A}{B}\right) = \ln A - \ln B$ }

Sol3 → MLE for μ

* Maximize $l(\mu,\sigma)$ w.r.t μ

$$l(\mu,\sigma) = \underbrace{\boxed{-n\ln(\sqrt{2}\sigma)}}_{\text{not dependent on } \mu} - \frac{\sqrt{2}}{\sigma}\sum_{i=1}^{n}|x_i-\mu|$$

* So, To Maximize $l(\mu,\sigma)$, we need to minimize $\sum_{i=1}^{n}|x_i-\mu|$, Since the negative sign means smaller $\sum|x_i-\mu|$ leads to larger likelihood.

* So, finding MLE for μ is equivalent to

$$\hat{\mu} = \underset{\mu}{\operatorname{argmin}} \sum_{i=1}^{n}|x_i-\mu|$$

This means μ should minimize the sum of absolute deviation (errors) between the data points x_i & μ.

So, MEDIAN is the value of μ that minimizes the sum $\sum|x_i-\mu|$.

Intuition → If μ is too far from the center of the data, the sum $\sum|x_i-\mu|$ becomes larger because the deviation $|x_i-\mu|$ are larger.

* for the data $\{-2,-1,0,2,3\}$, $\boxed{\text{the median is } \hat{\mu}=0}$

Sol4 → MLE for σ →

Differentiate l w.r.t σ

$$\frac{\partial l(\mu,\sigma)}{\partial\sigma} = \frac{\partial}{\partial\sigma}\left[-n\ln(\underbrace{\sqrt{2}}_{A}\underbrace{\sigma}_{B})\right] - \partial\left(\frac{\sqrt{2}}{\sigma}\sum_{i=1}^{n}|x_i-\mu|\right)$$

$$= -n\frac{\partial}{\partial\sigma}\left[\ln(\sqrt{2})+\ln(\sigma)\right] - \frac{\sqrt{2}}{1}\frac{\partial}{\partial\sigma}\left[\frac{\sum_{i=1}^{n}|x_i-\mu|}{\sigma}\right]$$

$$= -n\left[0+\frac{1}{\sigma}\right] - \underbrace{\frac{\sqrt{2}}{1}\sum_{i=1}^{n}|x_i-\mu|}_{\text{constant}}\frac{\partial}{\partial\sigma}\left(\frac{1}{\sigma}\right)$$

Sol 4 (continue)

$$\frac{\partial l(\mu,\sigma)}{\partial \sigma} = \frac{-n}{\sigma} - \sqrt{2}\sum_{i=1}^{n}|x_i-\mu|\left(\frac{-1}{\sigma^2}\right)$$

$$= \frac{-n}{\sigma} + \frac{\sqrt{2}\sum_{i=1}^{n}|x_i-\mu|}{\sigma^2}$$

Reference

$$\frac{d}{dx}\left(\frac{1}{x}\right) \Rightarrow \frac{d}{dx}(x^{-1})$$

$$\Downarrow$$

$$-\frac{1}{x^2} \Leftarrow -x^{-2}$$

Set $\frac{\partial l(\mu,\sigma)}{\partial \sigma} = 0$ to maximize the likelihood

* $\frac{-n}{\sigma} + \frac{\sqrt{2}}{\sigma^2}\sum_{i=1}^{n}|x_i-\mu| = 0$

* $\frac{-n\sigma + \sqrt{2}\sum_{i=1}^{n}|x_i-\mu|}{\sigma^2} = 0$

* $-n\sigma = -\sqrt{2}\sum_{i=1}^{n}|x_i-\mu|$

* $\boxed{\sigma = \frac{\sqrt{2}\sum_{i=1}^{n}|x_i-\mu|}{n}}$

for data $\{-2,-1,0,2,3\}$ & $\mu = 0$ & $n=5$ (no. of data points)

$$\sigma = \frac{\sqrt{2}}{5}\left[|-2-0|+|-1-0|+|0-0|+|2-0|+|3-0|\right]$$

$$\sigma = \frac{\sqrt{2}}{5}[2+1+0+2+3] = \frac{\sqrt{2}}{5}\times 8$$

$$\boxed{\sigma = \frac{8\sqrt{2}}{5}}$$

$$\boxed{\text{final Answer} \rightarrow \mu = 0 \ \& \ \sigma = \frac{8\sqrt{2}}{5}}$$

1) log likelihood function is monotonic transformation of L, maximizing l is same as maximizing L.